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A MULTILEVEL INVESTIGATION OF ANTECEDENTS AND CONSEQUENCES OF TEAM MEMBER BOUNDARY-SPANNING BEHAVIOR

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We adopt a multilevel approach to a paradox that arises when research findings on the performance benefits of team boundary spanning are juxtaposed with earlier work demonstrating the role overload of individuals who span boundaries. Results revealed that individual and team-level factors predicted member boundary-spanning behavior, which increases individual role overload, which then negatively impacts team viability. However, when we aggregated to the team level of analysis, higher levels of boundary spanning resulted in team members' experiencing significantly less role overload. Implications for addressing the boundary-spanning paradox and developing a multilevel theory of team boundary spanning are discussed.

In an era of flatter, network-based organizational designs (Choi, 2002), multiteam systems (Marks, DeChurch, Mathieu, Panzer, & Alonso, 2005), and continuous innovation (Hargadon, 1998), teams are increasingly responsible for completing complex cross-functional tasks and bridging organizations to create and transfer valuable sources of knowledge and know-how (Ancona & Caldwell, 1992). In order to meet such demands, teams such as those working in product design, consulting, and strategic alliances must directly interface with important stakeholders both within and outside of their organization (Ancona, Bresman, & Kaeufer, 2002). These externally oriented activities include managing changing customer requirements, negotiating project scope, and acquiring key sources of information from external parties and are generally referred to as *boundary spanning* or *boundary management* (Ancona, 1990). Effective boundary management not only directly benefits team effectiveness, but also—since team boundaries are important for knowledge transfer within and between organizations (Argote, McEvily, & Reagans, 2003; Hansen, 1999), for organizational innovation (Hargadon, 1998), and for protection against outside threats (Aldrich & Herker, 1976)—translates into improved organizational learning (Edmondson, 1999) and effectiveness (Carlisle, 2004).

To date, most scholarly work exploring boundary spanning has been conducted within two separate streams of research, each concentrating at a different level of analysis and remaining largely unconnected to the other. The more recent stream, centered on the team level, has identified strategies that teams can use to manage their external environments and ways in which these boundary-spanning activities facilitate team effectiveness (e.g., Ancona, 1990; Ancona & Caldwell, 1992). In contrast, earlier findings from social-psychological research conducted at the individual level have demonstrated that those who carry out boundary-spanning responsibilities, although gaining status and influence through access to unique knowledge, also experience significant role overload as a result of facing simultaneous and often conflicting pressures (Kahn, Wolfe, Quinn, Snoek, & Rosenthal, 1964; Katz & Kahn, 1978). Juxtaposing these seemingly contradictory sets of research findings raises an interesting and practically significant potential paradox: a team engaging in boundary spanning may more effectively manage its external environment in ways that aid performance, yet, in carrying

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out boundary spanning, team members may experience significant role overload that could harm the team's long-term viability. Given the need for a team to both perform well and to build and maintain members' capabilities for working as a team over time in ways that promote learning and innovation (Hackman, 1987; Sundstrom, DeMuese, & Futrell, 1990), addressing this paradox has important implications.

Accordingly, the purpose of our study was to develop and test a multilevel model of team boundary spanning that would enable us to investigate and understand a potential resolution to this paradox. Because boundary spanning is critical to team performance but also requires considerable effort and time from team members, understanding the paradox begins with investigating the conditions that motivate such behavior. Teams depend on their members to perform boundary-spanning activities that are important for team performance, yet many team members underengage in these externally directed activities and instead focus on their teams' internal activities and processes (Ancona, 1990; Ancona & Caldwell, 1992). By adopting a multilevel approach aggregating boundary spanning measured at an individual level to the team level of analysis, we make the contribution of identifying factors that influence team member engagement in these critical yet challenging behaviors. Specifically, we examine the interplay among individual-level characteristics, such as perceived role responsibilities and confidence in one's boundary-spanning ability, and team-level factors, such as the degree to which a team has an explicit external focus and team members agree on that focus. A better understanding of the mechanisms supporting individual boundary-spanning behavior helps to answer recent calls to identify the antecedents of team boundary spanning (Choi, 2002; Edmondson, 1999) and advances selection and development efforts aimed at improving team boundary management.

A second contribution made here is investigating the team boundary spanning paradox itself and connecting two previously disparate but potentially complementary literatures. Through examining the outcomes of boundary spanning at the team and individual levels and their interrelationships, we consider how team performance might benefit from boundary spanning while minimizing individual costs that diminish long-term team viability. This approach will help to better explain outcomes at each level and provide a more integrated understanding of seemingly divergent outcomes of boundary spanning at different levels. A richer understanding of the conditions that lead to improved team boundary management offers important implications not only for teams and team members, but also for organizations relying on teams to facilitate key knowledge and resource flows within and outside organizational boundaries (Argote et al., 2003).

THEORY AND HYPOTHESIS DEVELOPMENT

Team Member Boundary Spanning

In keeping with prior work (e.g., Ancona, 1990; Ancona & Caldwell, 1992), we define team members' boundary spanning as behaviors intended to establish relationships and interactions with external actors that can assist their team in meeting its overall objectives. Several points are important to note. First, boundary spanning includes a range of externally directed actions. Certain behaviors serve an information-gathering function, as team members seek, interpret, and communicate information from external contacts (e.g., Hansen, 1999), whereas other behaviors fulfill an external representation function, as team members communicate with external stakeholders so as to enable their team to set expectations on team objectives, frame requests for needed resources, and update managers on project status, all of which legitimatize the team and/or buffer outside pressures (cf. Aldrich & Herker, 1976). Second, team member boundary spanning involves connecting to important external actors, who may include key clients, managers or senior executives, and other individuals or groups with whom the team works interdependently or who can provide valued and needed resources. Third, these behaviors fit the definition of external processes noted in team effectiveness models (e.g., Gladstein, 1984; Sundstrom et al., 1990) and are distinct from internal team processes, such as conflict management, member coordination, and goal setting (Gladstein, 1984).

Although prior research on boundary spanning has focused on demonstrating its contributions to performance (e.g., Ancona & Caldwell, 1992), less attention has been given to understanding potential antecedents (Choi, 2002). Below, we focus on individual-level and team-level predictors that motivate team members to carry out the challenging activities comprising boundary spanning. Our approach to boundary spanning at the team level of analysis is to represent it as the additive compilation of team members' boundary-spanning behaviors (Chan, 1998). Since we frame boundary spanning as originating at the individual level of analysis, we focus on what we believe to be important predictors of the degree to which team members engage in boundary-spanning activities. Our hypothesized model, which is presented in Figure 1, applies most directly to teams for which externally directed activities are particularly important. Specifically, these would be teams that (1) work in relatively uncertain and demanding environments with diverse constituents, (2) have high interdependence with outside individuals and groups, and (3) experience high task complexity (Ancona, 1990; Choi, 2002).

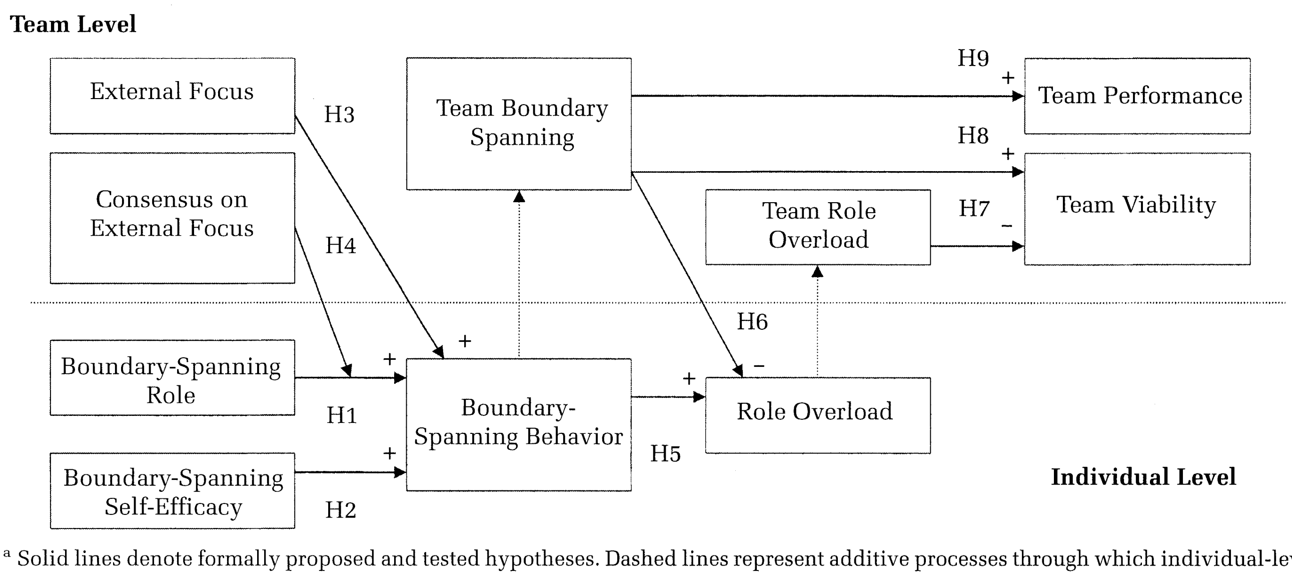
Individual-Level Antecedents of Boundary-Spanning Behavior

One point on which previous research is clear is that boundary-spanning activities are taxing and require a great deal of effort and persistence (e.g., Ancona, 1990; Kahn et al., 1964). Two individual-level factors may be particularly important in light of these specific challenges. First, boundary-spanning roles are expected to provide a structure that defines and directs team member efforts and reinforces members' boundary-spanning responsibilities (cf. Katz & Kahn, 1978). Second, boundary-spanning self-efficacy should provide team members with the necessary confidence in their own capabilities to successfully carry out boundary-spanning tasks (cf. Bandura, 1997). Boundary-spanning roles and self-efficacy are each important because of their potential predictive value, and they are also likely to be of practical significance, since both factors can be considered and actively managed through team composition and design.

Boundary-spanning roles. Team member roles are generally conceptualized as socially constructed frameworks of behaviors and actions that are appropriate and expected (Ilgen & Hollenbeck, 1991). Because boundary-spanning activities are taxing and can directly compete with internally directed activities for time and attention (Choi, 2002; Katz & Kahn, 1978), team members are unlikely to engage in these external activities unless they believe them to be a part of their personal role responsibilities. Role theory research (e.g., Katz & Kahn, 1978) suggests that team members develop expectations about each others' roles that guide beliefs about desired and appropriate role behaviors. Consequently, team members often provide direct indications about rewards and sanctions associated with role compliance in an effort to influence role conformity and facilitate the accomplishment of team objectives (Hackman, 1992). In addition to external pressures and social cues regarding the instrumentality of engaging in role behaviors, those assuming boundary-spanning roles experience internal pressures to conform to expectations and gain social approval (Blau, 1960). Other internal motivators may include intrinsic satisfaction from fulfilling challenging roles and/or successfully demonstrating valued role performance (Katz & Kahn, 1978). We hypothesize:

Hypothesis 1. The extent to which team members assume a boundary-spanning role is positively related to the degree to which they engage in boundary-spanning behavior.

FIGURE 1
Hypothesized Multilevel Model of the Antecedents and Consequences of Boundary Spanning[a]



[a] Solid lines denote formally proposed and tested hypotheses. Dashed lines represent additive processes through which individual-level phenomena are compiled to form team-level phenomena.

Boundary-spanning self-efficacy. Team members need to feel confident that they can perform such challenging and time-consuming activities in conjunction with their other teamwork and task responsibilities. Given the challenge and complexity of boundary-spanning behaviors and the influence self-efficacy has on adaptation and persistence during difficult and complex tasks (Bandura, 1997), a team member's boundary-spanning self-efficacy should be particularly important. Following Bandura's (1997) conceptualization of self-efficacy, we define *boundary-spanning self-efficacy* as an individual's confidence in his or her ability to successfully establish and manage relationships with important parties external to his or her team. Highly efficacious individuals feel confident interacting with external parties and believe that they have the necessary skills to effectively manage a variety of external variables impacting the team. Because of their confidence, these members are not only more likely to take on boundary-spanning tasks, but also are more likely to control the negative and otherwise harmful emotional reactions that often arise during challenging and taxing situations; they are also more likely to perceive the demands and challenges associated with boundary spanning as opportunities to excel rather than obstacles to avoid (Bandura, 1997). Consequently, team members with greater self-efficacy will set higher personal goals with respect to boundary spanning and will be more likely to generate task-appropriate strategies for carrying out these challenging behaviors successfully (Locke & Latham, 1990). On the basis of the foregoing, we predict:

Hypothesis 2. Boundary-spanning self-efficacy is positively related to team member boundary-spanning behavior.

Team-Level Antecedents of Individual-Level Boundary Spanning

Teams provide important contexts for influencing team member behaviors (Hackman, 1992). In the boundary-spanning domain, Ancona's qualitative work showed that when teams consciously and purposefully considered externally oriented strategies and did so in a way that ensured that all team members understood the importance of boundary-spanning activities, they were more successful at boundary management. This finding suggests not only that it might be important for teams to be strategic and purposeful in managing their environments, but also that their members should have consistent and high levels of agreement regarding external focus.

External focus. Reflecting team members' collective perceptions about the relevance and value of boundary spanning as a critical team function, external focus has been identified as important for teams that are highly interdependent with external constituents (Choi, 2002). During initial planning meetings, team members engage in a variety of strategic internal processes, such as identifying the team mission and overall objectives and formulating team strategies (Marks, Mathieu, & Zaccaro, 2001). Teams that collectively value boundary-spanning activities should be more likely to incorporate them into their team strategy, monitor these activities accordingly, and actively encourage boundary-spanning behaviors throughout the team's period of operation.

After studying newly formed consulting and research and development teams, Ancona and colleagues found that teams significantly differed in their strategies for engaging and managing their external environments (Ancona, 1990; Ancona & Caldwell, 1992). Teams following strategies of high visibility and extensive external activity engaged in high levels of boundary spanning and significantly outperformed teams utilizing either mixed or predominately inward-focused strategies. Choi (2002) similarly theorized that teams likely differ on the degree to which they mindfully adopt externally oriented strategies. Overall, this work suggests that the level of strategic emphasis placed on external activities varies across teams and that these orientations may influence the degree to which team members undertake boundary spanning. Additional support comes from Hoegl, Parboteeah, and Munson (2003), who studied 145 software development teams and found that collective perceptions of the necessity of interacting with team-external contacts to acquire knowledge, work contributions, or feedback predicted team members' engagement in actual network-building behavior. Thus, we posit:

Hypothesis 3. The level of a team's external focus is positively related to individual boundary-spanning behavior.

Moderating role of consensus on external focus. Researchers have recently argued that consensus among a team's members on the team's strategic imperatives is important for shaping processes and motivating behaviors that facilitate goal accomplishment (Colquitt, Noe, & Jackson, 2002; Kozlowski & Ilgen, 2006). Indeed, scholars have suggested the importance of studying team members' formation of a collective understanding of their external requirements and the similarity in these perceptions (Choi, 2002; Edmondson, 1999). Because boundary-spanning activities require considerable effort and time, team members who assume externally directed roles may be more likely to actually carry out these activities if there is team consensus on their importance. Specifically, consensus among members on the importance of an external focus should influence the extent to which a team sends consistent and clear messages to its members regarding the value and desirability of boundary-spanning activities.

According to role theory (Katz & Kahn, 1978), consensus on the value of boundary spanning is a necessary prerequisite for sending strong and clear signals regarding behavioral expectations to those in boundary-spanning roles. The greater the clarity and strength of "sent role expectations," the less likely role messages will be misunderstood or distorted; thus, clear expectations strengthen the relationship between a team member's assuming a boundary-spanning role and his or her engaging in boundary-spanning behavior. Furthermore, other team members will be more likely to evaluate and monitor compliance with role expectations if they agree strongly that a failure to maintain external relationships would be costly to the team (Aldrich & Herker, 1976). Examples of such monitoring may include direct messages of approval or disapproval regarding role performance and frequent requests about the status of boundary-spanning activities at team meetings. In these ways, consensus with respect to external focus serves as a strong discretionary stimulus in that it is "transmitted or made available to individuals differentially and selectively at the discretion of other group members" (Hackman, 1992: 201). In this case, strong and consistent expectations will be directed specifically at those team members assuming boundary-spanning roles. We expect, then, that the positive relationship between assuming a boundary-spanning role and engaging in boundary-spanning behavior will be strongest when the degree of agreement among team members on the importance of boundary spanning is high. Stated formally:

Hypothesis 4. Consensus on team external focus serves as a cross-level moderator: the positive relationship between boundary-spanning roles and team member boundary-spanning behavior is stronger in teams with a strong consensus on an external focus than in teams with a weak consensus.

Consequences of Boundary Spanning at the Individual and Team Levels of Analysis

Role overload. Despite the purported benefits of boundary spanning for teams, boundary spanning is stressful and challenging, requiring considerable effort and time (Aldrich & Herker, 1976), and these characteristics likely contribute to role overload. Role overload, which exists when an individual has too much work to do in the time available (Beehr, Walsh, & Taber, 1976), is a type of person-role conflict in which the individual must decide on which activities to do and which to delay (Kahn et al., 1964). These costs in terms of effort and time are largely attributable to the fact that those engaging in boundary spanning are responsible for actively managing a variety of internal and environmental contingencies either simultaneously or sequentially (Katz & Kahn, 1978). Because carrying out these externally directed behaviors requires the boundary spanners to be connected both internally to their teams and externally to outsiders, these efforts naturally require spanning different and sometimes conflicting subsystems (Aldrich & Herker, 1976; Katz & Kahn, 1978), which can further contribute to role overload. Additionally, because boundary spanning requires a balance with internally focused processes (Ancona, 1990; Choi, 2002), team members may find that they have significantly greater internal role demands as they engage in additional boundary spanning. This can generate role overload by increasing the total number of role demands a team member experiences. We predict:

Hypothesis 5. Individual-level boundary spanning is positively related to the amount of individual role overload that individuals experience.

Although the evidence presented above suggests that when team members take on high levels of boundary spanning, they will experience greater role overload, there are also conceptual reasons for suggesting that individual role overload may be *reduced* when teams as a whole engage in significant boundary spanning (Choi, 2002). Specifically, building from previous research by Ancona and colleagues demonstrating the gains associated with external team processes (e.g., Ancona, 1990; Ancona & Caldwell, 1992; Gladstein, 1984), team boundary spanning should help teams by reducing the degree to which individual team members experience role overload. In particular, ambassadorial activities (Ancona, 1990; Ancona & Caldwell, 1992) protect teams against unreasonable external expectations and team members against information overload, difficult demands and requests, and/or impossible project deadlines. Such protection can help ensure that team members are not unduly overloaded as they work toward accomplishing team objectives. In addition, when teams "boundary span," more knowledge and information of clients, managers, and other important external parties is potentially available to team members, which can help them manage task strategies, project deadlines, and work coordination in ways that can reduce team members' stress and role overload (Choi, 2002). Further, teams engaging in higher levels of boundary spanning receive more support from critical external constituencies such as senior managers, perhaps in the form of resources that may help team members manage their team responsibilities and obligations and help minimize team members' role overload. Thus, we hypothesize a direct and negative relationship between team-level boundary spanning and team member role overload that goes above and beyond the influence of individual-level boundary spanning.

Hypothesis 6. Team-level boundary spanning is negatively related to the amount of individual role overload that individuals experience.

Team viability. Team effectiveness is multidimensional and consists not only of team performance as determined through the quantity, quality, or end user satisfaction with a team's output, but also includes the team's ability to function interdependently in the future (Hackman, 1987; Sundstrom et al., 1990). Although prior research on boundary spanning has focused on team performance (e.g., Ancona & Caldwell, 1992), considering team viability in the context of boundary spanning is important given that externally directed boundary-spanning activities can directly compete for members' attention and effort against internal processes that develop and maintain team identity and cohesion (Ancona, 1990; Choi, 2002). To explore this further, we highlight two simultaneously operating mechanisms, each of which exerts distinct and conflicting influences on team viability. On the one hand, the benefits of boundary spanning at the team level, such as receiving valued resources and social support from external sources (e.g., Ancona, 1990), may help to cultivate viability over time, but on the other hand, the challenges and overload experienced by individual team members carrying out these behaviors (e.g., Beehr et al., 1976) may foster an adverse internal dynamic that ultimately reduces a team's viability. We discuss both of these perspectives below.

When a team's members experience role overload, they are less satisfied with their team experience and less likely to look forward to future opportunities to work together, thereby making the team less viable (Hackman, 1987; Sundstrom et al., 1990). Specifically, team members who are more stressed and overloaded are likely to feel frustrated and anxious, and they are less likely to help other team members, thereby creating an overall unpleasant and dissatisfying team experience, which likely decreases a team's ability to retain its members over time (Gersick & Hackman, 1990). Levi (2001) noted that role stressors are especially likely to become salient and detrimental as a team approaches the end of its project term. If team members are overloaded and struggling to complete assigned tasks, they may become increasingly aware of the different expectations members have about who will do what and what roles each individual should be performing. Uncovering these divergent expectations, coupled with the stress of being overloaded and facing imminent deadlines, can lead to dysfunctional conflict and decreased team morale. Thus, we expect that when the members of a team experience high levels of role conflict overall, the team will have lower viability. Stated formally:

Hypothesis 7. Team-level role overload is negatively related to team viability.

Team boundary spanning may, however, support team viability. The collective boundary-spanning efforts of team members at the team level enable teams to acquire needed resources, support, and guidance from external parties, which should in turn promote a highly productive internal team dynamic and a satisfying team experience overall. Although the impact of team boundary spanning on team viability has largely been unexplored, Ancona (1990) did provide some evidence that, over time, high levels of individual contentment and team cohesion were reported in teams that actively engaged their external environments. Qualitative data revealed that the members of those teams that interacted regularly with key external clients and other parties were cognizant of and highly satisfied with their teams' ability to gain pertinent information from key outsiders and manage project objectives accordingly. These team members also reported a strong internal "spirit" and a desire to stick together. In line with these findings, we posit that:

Hypothesis 8. Team-level boundary spanning is positively related to team viability.

Team performance. Finally, we expect team-level boundary spanning to be positively related to team performance. Existing studies of team boundary spanning support this assertion. For instance, Gladstein (1984) found that external team processes (but not internal processes) were significantly and positively related to objective ratings of team performance. In a later study, Ancona and Caldwell (1992) reported that new-product development teams engaging in a variety of boundary-spanning activities were significantly more innovative over time and were also more successful at meeting schedules and budgets because they were better able to effectively identify and manage senior executives' expectations, buffer debilitating external pressures, and acquire key outside information and guidance. Finally, in a study of education consulting teams, Ancona (1990) found that engaging in boundary spanning helped teams better understand client demands and needs, which improved delivery of services and promoted the teams' achievements to key stakeholders. Thus, we hypothesize:

Hypothesis 9. Team-level boundary spanning is positively related to team performance.

METHODS

Sample and Data Collection

We collected data from a sample of 190 MBA students, comprising 31 consulting teams (5–7 members per team), at a large mid-Atlantic university. The sample was a realistic and appropriate research setting for exploring boundary spanning as the teams were highly reliant on external sources for carrying out their work. Every second-year MBA student was required to participate in an intensive, semester-long, actual consulting engagement with a real organization. Participants were assigned to consulting teams on the basis of their concentration areas and domains of experience. Teams were engaged to address specific business needs, faced tight time pressures and aggressive deadlines, and delivered final recommendations that their clients evaluated and implemented as they deemed appropriate. These conditions provided challenging and highly realistic consulting experiences. Further, this sample was well suited to testing our hypotheses because teams had very similar tasks and life cycles, thus ruling out confounds arising from differences in these factors in our empirical analysis.

We assessed constructs at times during the team life cycle that were appropriate to the underlying causal model and minimized common method variance (Podsakoff, MacKenzie, Jeong-Yeon, & Podsakoff, 2003). At approximately 4 weeks into the projects, Survey 1 captured team member boundary-spanning self-efficacy and team member perceptions of the value and relevance of boundary spanning as a critical team function (an external focus). At approximately 11.5 weeks, Survey 2 provided data on boundary-spanning role responsibilities and behaviors. Finally, we administered Survey 3 to all team members at project completion (final deliverable presentations to clients) to obtain supplemental information regarding challenges and obstacles faced during the projects. This information was utilized as a control variable in our tests of hypotheses. We also surveyed the clients at this time to obtain independent measures of team performance. Surveys were received from 190 team members (a 100 percent response rate) and from clients of 30 of the 31 teams (a 97 percent response rate).

Measures

Boundary-spanning behavior. Closely following existing research, we measured individual boundary-spanning behavior with six items adapted directly from Ancona and Caldwell's (1992) study and modified to reflect the current consulting team context. Team members rated the degree to which each of their teammates engaged in boundary-spanning behaviors (1, "not at all," to 5, "to a very great extent"). Peer ratings were appropriate because team members worked together frequently, often daily, and thus had ample opportunity to witness and evaluate behavior. Items had the stem "To what extent did this team member . . ." and included "persuade outsiders (e.g., faculty, clients) to support team decisions," "prevent the outsiders from 'overloading' the team with too many requests," "reach out to individuals outside of your team that can provide project-related expertise or ideas," and "proactively seek the advice and support of your faculty advisor." A principal components analysis yielded a single factor (α = .86). Interrater and intraclass measures (median r_{wg} = .80, ICC[1] = .28, ICC[2] = .71) justified aggregation across raters (James, Demaree, & Wolf, 1984; Shrout & Fleiss, 1979).

Boundary-spanning roles. In accordance with role theory, we had all participants rate the extent to which they perceived each of the six boundary-spanning behaviors as an expected part of their responsibilities (1, "not at all," to 5, "to a very great extent"). Principal components analysis yielded a single factor (α = .86).

Boundary-spanning self-efficacy. We measured boundary-spanning self-efficacy by asking team members to rate their confidence in their ability to successfully engage in a variety of boundary-spanning behaviors. Scale items were based in part upon Parker's (1998) role breadth self-efficacy scale, which asks respondents to rate how confident they feel about contacting individuals outside of their own work units. Sample items, with the stem "Based upon my past experiences working in teams, in my MBA consulting team, I feel very confident. . . ," included "establishing a good rapport with key stakeholders external to the team" and "representing my team to other external parties as we discuss project business." Eight items were rated on a five-point disagree/agree Likert scale, and principal components analysis yielded a single factor (α = .92).

External focus. Drawing on Ancona (1990) and Choi (2002), we asked team members to rate the extent to which their team saw boundary-spanning activities as important. The level of external focus was defined as the proximity of the group mean to the positive endpoint of this response scale. Items had the stem "My MBA consulting team places great importance on . . ." and included "building solid relationships with key external stakeholders (faculty advisors and clients)," "acquiring knowledge for this project from persons external to the team," and "collaborating with other professionals (outside of our team) that can offer support and guidance." The four items were rated on a five-point disagree/agree scale, and principal components analysis yielded a single factor (α = .91). The measure of agreement among team members' ratings produced a median r_{wg} of .66, an ICC(1) of .09, and an ICC(2) of .38. Because these aggregation statistics were either within or only slightly below the acceptable range of values summarized in the literature (e.g., Bliese, 2000; James, 1982) and comparable with other previously reported values in specific studies (e.g., Schneider, White, & Paul, 1998), they did not seem low enough to prohibit aggregation, particularly when they viewed in combination, as suggested by Bliese (2000). Moreover, conceptually, prior work supports the existence of team-level perceptions of the importance of boundary-spanning functions for teams (Ancona, 1990; Ancona & Caldwell, 1992; Choi, 2002). Additionally, these values suggested that although significant variability existed in members' responses as a function of team membership, there was also sizable variance within teams in members' beliefs about the importance their team placed on externally directed activities. These patterns of variance support our use of consensus below.

Consensus on external focus. We defined consensus on external focus as the dispersion of respondents' ratings of external focus. Teams with a stronger consensus on external focus had members who agreed more completely with one another on the importance of an external focus. Following Chan's (1998) dispersion model and prior examples of team consensus measures (e.g., Colquitt et al., 2002), we operationalized this variable as the standard deviation across all team members on the four external focus items, thus capturing the extent of agreement, or consensus, within each team. Lower standard deviations reflect higher within-team agreement.

Role overload. Three items were adapted from Beehr et al. (1976) and modified for the current context. Items had the stem "When it comes to my roles and responsibilities on this team . . ." and were "it often seems like I have too much work for one person to do," "the expectations for what I should do are too high," and "I feel that I have taken on too much." They were rated on a five-point disagree/agree Likert scale. A principal components analysis yielded a single factor (α = .83).

Team role overload. Following an additive compositional model logic (Chan, 1998), we calculated role overload at the team level as the aggregated mean score for all team members on each of the three role overload items noted above. Because the team role overload score conceptually and empirically reflected the simple average calculated across all members, empirical justification for aggregation was not necessary (Chan, 1998).

Team boundary spanning. Following Ancona and Caldwell's (1992) measurement approach, and again in accordance with an additive compositional model (Chan, 1998), we calculated the team boundary-spanning behavior variable as the aggregated mean score for all team members on each of the six boundary-spanning behavior items. Because this variable conceptually and empirically reflected a simple average, again, justification of aggregation was not necessary.

Team viability. Team viability, or the extent to which a team was able to increase its ability to perform as an intact unit over time, was assessed via three items developed for this study (based largely on Hackman [1987] and Sundstrom et al. [1990]). Following Chan's (1998) referent-shift consensus model, we used a team-level referent. Members used a five-point disagree/agree scale to rate the extent to which their team provided a satisfying and developmental experience and could continue working productively over time. Items were "Team members have found being a member of this team to be a very satisfying experience," "Most team members feel like they are learning a great deal by working on this project," and "Most of the members of this team would welcome the opportunity to work as a group again in the future" (median r_{wg} = .78, ICC[1] =.59, and ICC[2] = .74; single factor; α = .81).

Team performance. Client contacts who had worked closely with a team and were the end users of its project's results rated team performance. Items included "meeting specified project deadlines in a timely manner," "providing recommendations that are feasible (i.e., can realistically be implemented)," "successfully managing project-related challenges or obstacles as they occurred," and "managed their time with you and other members of your company effectively." The first two items were rated on a scale ranging from 1, "extremely ineffective," to 7, "extremely effective," and the second two items were rated on a scale ranging from 1, "strongly disagree," to 7, "strongly agree." We combined the four items to create an overall scale. Principal components analysis yielded a single factor (α = .84).

Control variables.[1] Previous research has shown effects among demographic attributes and employee attitudes and behaviors (Williams & O'Reilly, 1998). Consequently, we included *Asian ethnicity* (1, "Asian," and 0, "other"), and *GMAT score* (a continuous score) as controls when predicting individual boundary-spanning behavior. Additionally, *team size,* a fixed resource that may influence individuals' ability to carry out specific behaviors, was included as control variable. Finally, extensive challenges and/or high levels of *project demands* may also restrict a team's ability to carry out specific behaviors or influence the importance of boundary spanning (Choi, 2002). Consequently, we asked team members to rate two items indicating the degree to which their team's project involved an "extremely high level of client demands/expectations related to the project" and the "challenging nature of the project itself (e.g., complex technical requirements)" using a scale ranging from 1, "not present at all," to 7, "very much present" (single factor; α = .76; median r_{wg} = .82, ICC[1] = .45; and ICC[2] = .84).

Analyses

Because data were nested (i.e., individuals were nested within teams), we utilized hierarchical linear modeling (HLM) to test certain hypotheses. HLM is a statistical technique allowing researchers to examine relationships across multiple levels of analysis through a two-level approach that takes into account nonindependence inherent within nested data by simultaneously partitioning and modeling within-group and between-group variance (Raudenbush & Bryk, 2002). First, for each dependent variable of interest, we ran a set of *null models* with no predictors. Resulting ICC(1) values reflecting the percentage of variance residing between groups indicated HLM analyses were appropriate.[2] We then used level 1 analyses to determine the significance of hypothesized individual-level antecedents in predicting team member outcomes. Level 2 analyses followed, to which we added team-level antecedents to test for additional explained variance.

RESULTS

Table 1 presents descriptive statistics and bivariate correlations for all study variables. We should note that correlations at the individual level do not take into account nonindependence within the data and therefore should be interpreted cautiously.

Results of HLM Null Models

Null models were run for the two individual-level dependent variables of interest. Resulting ICC(1) values and associated chi-square tests revealed that 31 percent of the variance in team member boundary-spanning behavior resided between groups (χ^2[30] = 111.36, $p < .001$), and 18 percent of the variance in team member role overload resided between groups (χ^2[30] = 69.20, $p < .001$). Accordingly, we used HLM to predict team member boundary-spanning behavior (Hypotheses 1–4) and role overload (Hypotheses 5 and 6).

Antecedents of Team Member Boundary-Spanning Behavior

Table 2 summarizes the results of HLM analyses testing Hypotheses 1–4.

Hypothesis 1 posits that the extent of a team member's boundary-spanning role positively predicts engagement in boundary-spanning behavior,

[1] We initially explored participant gender, age, functional background, match of functional background with consulting task domain, cognitive ability, ethnic background, perceptions of project value at the individual level, team size, team project demands, team diversity (gender, age, ethnicity), familiarity of team members, task functional domain, task interdependence, need for boundary spanning, and advisor boundary-spanning behavior at the team level. Only Asian ethnicity, GMAT score, team size, and project demands were found to be significantly correlated with the dependent variables of interest. Therefore, to maximize power, we included only these four variables as controls in subsequent tests of hypotheses. As an additional check, we also reran all the HLM analyses and ordinary least squares (OLS) regression analyses used for hypothesis testing to see if including any of the larger set of potential control variables affected the results; none did so substantially.

[2] If little or no variance in the dependent variable resides between groups, as indicated by an insignificant ICC(1) value, the assumptions of OLS regression techniques are not violated and there is no need for HLM analyses.

TABLE 1
Descriptive Statistics and Correlations

Level 1 Variables	n	Mean	s.d.	1	2	3	4	5
1. Boundary-spanning behavior	190	2.91	0.70					
2. Boundary-spanning role	186	3.09	0.82	.40**				
3. Boundary-spanning self-efficacy	190	4.07	0.69	.25**	.25**			
4. Asian ethnicity	182	0.31	0.46	−.23**	.08	−.16*		
5. GMAT score	177	653.90	57.89	−.19*	−.05	−.18*	.25**	
6. Role overload	186	2.47	0.85	−.04	.14	−.02	.13	.01

Level 2 Variables	n	Mean	s.d.	1	2	3	4	5	6	7
1. Team boundary spanning	31	2.89	0.46							
2. External focus	31	3.37	0.44	.46**						
3. Consensus on external focus	31	1.07	0.25	−.29	−.51**					
4. Project demands	31	4.81	1.22	−.25	.21	−.17				
5. Team size	31	6.13	0.85	.26	−.01	−.00	−.14			
6. Team role overload	31	2.48	0.49	−.41*	−.33	.14	.40*	−.03		
7. Team viability	31	3.30	0.66	.37*	.54**	−.17	.30	.08	−.37*	
8. Team performance	30	5.81	1.14	.47**	.15	−.03	−.19	.13	−.41*	.38*

* $p < .05$
** $p < .01$

and Hypothesis 2 asserts that boundary-spanning self-efficacy predicts additional and unique variance in individual boundary-spanning behavior. We tested these hypotheses by entering the demographic controls (Asian ethnicity and GMAT score) and the two main effect variables (boundary-spanning roles and boundary-spanning self-efficacy) as level 1 predictors. Results supported both hypotheses: team members assuming boundary-spanning roles were significantly more likely to engage in boundary-spanning behavior than those who did not assume such roles ($\gamma = .22$, $p < .001$), and those with high levels of self-efficacy were significantly more likely to engage in boundary-spanning behavior than individuals with low self-efficacy ($\gamma = .15$, $p < .001$). As a block, the level 1 predictors explained 34 percent of the available within-group variance (69%) in boundary spanning ($R^2 = .34$).[3]

In Hypothesis 3, the degree to which teams emphasized externally oriented activities is expected to explain additional variance in boundary-spanning behavior. We entered two control variables (team size and project demands) and the main effect variable (external focus) as level 2 predictors, and we grand-mean-centered the level 1 predictors to test for incremental effects (Hoffman & Gavin, 1998) of these team-level variables going above and beyond the effects of the individual-level predictors identified in Hypotheses 1 and 2. Results supported Hypothesis 3 ($\gamma = .39$, $p < .05$) (Table 2). Team external focus explained an additional 17 percent of the available between-group variance (31%) in team member boundary spanning ($R^2 = .17$).[4]

Next, Hypothesis 4 predicts that within-team consensus on the importance of an external focus is a significant cross-level moderator of the relationship between roles and team member boundary-spanning behavior. To test this prediction, we added the variance variable as a level 2 predictor and as a predictor of the individual slope for boundary-spanning roles and boundary-spanning behavior, respectively. We group-mean-centered the level 1 predictors (Hoffman & Gavin, 1998) to avoid potential confounds when testing for cross-level moderation. Team consensus with respect to external focus did significantly moderate the relationship between roles and boundary-spanning behavior ($\gamma = -.45$, $p < .05$) (see Table 2). Hypothesis 4 was thus supported. Figure 2 illustrates the interaction.

As anticipated, the interaction plot revealed that

[3] The calculation was as follows: (*total within-group variance per the null model − residual within-group variance after considering level 1 predictors)/total within-group variance per the null model* (Bryk & Raudenbush, 1992).

[4] The calculation was as follows: (*total between-group variance per the null model − residual between-group variance after considering level 2 predictors)/total between-group variance per the null model* (Raudenbush & Bryk, 2002).

TABLE 2
Results of HLM Analyses Predicting Team Member Boundary-Spanning Behavior[a]

Variables[b]	Team Member Boundary-Spanning Behavior γ	t
Level 1 predictors		
Fixed effects		
Intercept	2.91	39.01***
Asian ethnicity	−0.28	−2.96**
GMAT score	−0.00	−1.52
Boundary-spanning role (Hypothesis 1)	0.22	4.19***
Boundary-spanning self-efficacy (Hypothesis 2)	0.15	3.22***
$R^2_{\text{within-group}}$	.34	
Level 2 predictors		
Fixed effects		
Team size	0.15	2.24*
Project demands	−0.15	−3.45**
External focus (Hypothesis 3)	0.39	2.07*
$R^2_{\text{between-group}}$	.17	
Consensus on external focus	−0.35	−1.02
Cross-level interactions		
Boundary-spanning role slope (Hypothesis 4)	−0.45	−2.19*

[a] Analyses were based on listwise deletion. $n = 190$, individual-level variables; $n = 31$, team-level variables.

[b] Level 1 predictor variables were group-mean-centered for testing level 1 effects, grand-mean-centered for testing incremental effects of level 2 predictors, and group-mean-centered for testing cross-level effects. We added consensus on external focus as a level 2 predictor in testing Hypothesis 4, as described in the text.

* $p < .05$
** $p < .01$
*** $p < .001$

the positive relationship between roles and boundary-spanning behavior was strongest when there was strong consensus on the importance of the boundary-spanning function. For teams with weak consensus, the positive slope between assuming a boundary-spanning role and engaging in boundary spanning was weaker.

Consequences of Boundary-Spanning Behavior

In Hypothesis 5, we state that team members engaging in more boundary spanning would experience more role overload, and in Hypothesis 6, we predict that the more boundary spanning that occurs at the team level, the less individual team members experience role overload. To test these hypotheses, we entered the control variables (Asian ethnicity and GMAT score) and team member boundary spanning as level 1 predictors. Results, presented in Table 3, support both hypotheses. Individual boundary-spanning behavior was significantly and positively related to role overload ($\gamma = .22$, $p < .05$) and boundary spanning by a team as a whole was significantly related to lower levels of team member role overload ($\gamma = -.66$, $p < .01$). Level 1 predictors explained 5 percent of the available within-group variance (72%) in role overload ($R^2 = .05$), and level 2 predictors explained 65 percent of the available between-group variance (18%) in role overload ($R^2 = .65$).

Because Hypotheses 7–9 concern team-level outcomes, they were tested via ordinary least squares regression. In Hypothesis 7, we predict that team role overload negatively impacts team viability. In Hypotheses 8 and 9, we assert that higher levels of team boundary spanning positively and significantly predict team viability and performance, respectively. The control variables (team size and project demands) were entered into step 1, followed by the predictor variable of interest in step 2 (team role overload for Hypothesis 7; team boundary spanning for Hypotheses 8 and 9). Table 4 presents results and shows that, as anticipated, team role overload significantly and negatively influenced team viability ($\beta = -.47$, $p < .05$). Team-level boundary-spanning behavior also positively predicted team viability ($\beta = .30$, $p < .10$) and team performance ($\beta = .44$, $p < .05$). Thus, Hypotheses 7, 8, and 9 all received support.

DISCUSSION

This study applied a multilevel theoretical lens and methodology as a means to integrate existing work on team boundary management with earlier research underscoring the challenges associated with carrying out boundary-spanning behaviors. Our results revealed two overarching findings: First, greater boundary spanning at the team level provided a positive and significant benefit to team members by reducing their role overload. This cross-level finding suggests that although boundary-spanning behaviors are taxing, challenging, and stressful for team members, when teams engage in high levels of boundary spanning these personal costs can be mitigated. Second, team members' boundary-spanning behaviors—which, when viewed in the aggregate, explain team effectiveness—are predicted by variables that reside at both the individual and team levels of analysis as well as by their interaction.

FIGURE 2
Cross-Level Moderating Effect of Consensus on External Focus on the Relationship between Team Member Role and Behavior

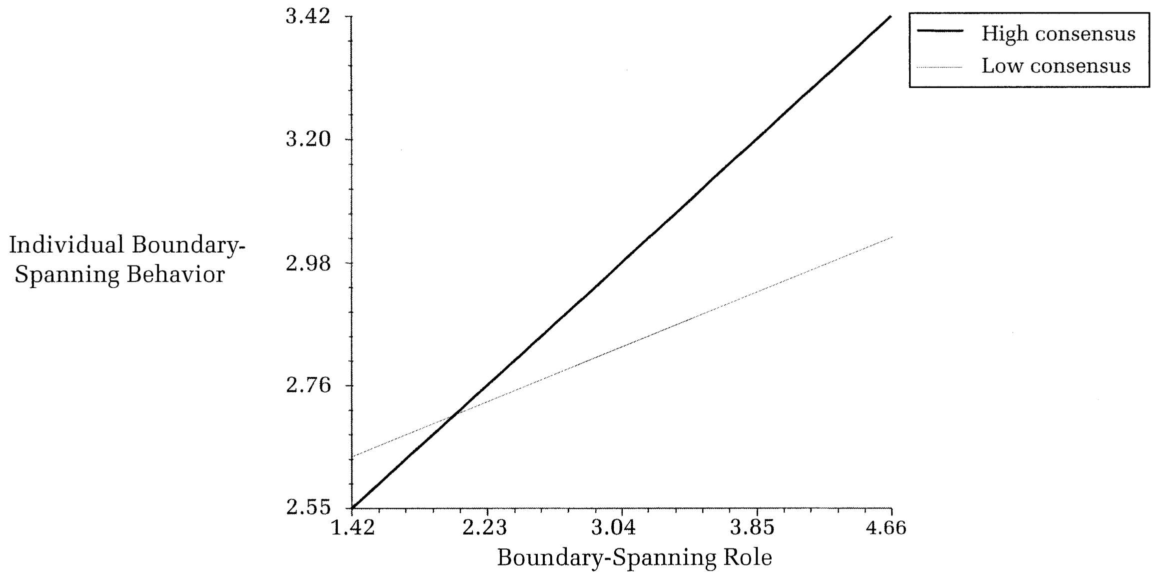


TABLE 3
Results of HLM Analyses Predicting Team Member Role Overload[a]

	Team Member Role Overload	
Variables[b]	γ	t
Level 1 predictors[b]		
Fixed effects		
Intercept	2.43	30.88***
Asian ethnicity	0.22	1.57
GMAT score	−0.00	−0.04
Boundary-spanning behavior (Hypothesis 5)	0.22	1.97*
$R^2_{\text{within-group}}$	.05	
Level 2 predictors		
Fixed effects		
Team size	0.15	1.64
Project demands	0.07	1.17
Team boundary spanning (Hypothesis 6)	−0.66	−3.31**
$R^2_{\text{between-group}}$	.65	

[a] Analyses were based on listwise deletion. n = 190, individual-level variables; n = 31, team-level variables.
[b] Level 1 predictors were group-mean-centered for testing level 1 effects and grand-mean-centered for testing incremental effects of level 2 predictors.
* $p < .05$
** $p < .01$
*** $p < .001$

Theoretical Contributions

Several theoretical contributions stem from these core findings and inform not only the team boundary spanning literature, but also the role theory, organizational learning, and knowledge management domains. First, we provide insights into a paradox that emerges when one integrates recent research on team-level benefits of boundary spanning with earlier work on the individual costs associated with carrying out boundary-spanning activities. Interestingly, although individuals working in teams routinely face such tensions, surprisingly little research directly explores these issues (Lewis, 2000). However, by integrating multiple levels of analysis, we delve deeper into the tradeoffs that exist across levels and offer a possible explanation. Specifically, the cross-level negative influence of team boundary spanning on individual role overload demonstrates that not only a team, but also the team members engaged in boundary spanning, fare better when others in the team are also engaging in boundary spanning.

Several possible mechanisms may explain this cross-level finding. For example, higher levels of team boundary spanning connect a team to more highly valued resources, such as information, feedback on progress, and support from key external parties (Ancona, 1990). These resource gains help the team accomplish its task and meet expectations, thereby reducing the stressors and demands

TABLE 4
Results of Regression Analyses for Team Effectiveness[a]

	Team Viability				Team Performance			
	Step 1		Step 2		Step 1		Step 2	
Variables	β	t	β	t	β	t	β	t
Step 1								
Team size	.13	0.70	.07	0.48	.11	0.58	.03	0.14
Project demands	.31	1.74[†]	.57	3.58*	−.18	−0.94	−.08	−0.46
Step 2								
Team role overload (Hypothesis 7)			−.47	−2.82*				
Team boundary spanning (Hypotheses 8 & 9)			.30	1.82[†]			.44	2.42*
R^2	.10		.46		.05		.22	
F	1.61		5.54*		0.69		2.50[†]	
ΔR^2			.36*				.18*	
ΔF			8.60*				5.86*	

[a] Analyses were based on listwise deletion. Estimates standardized regression coefficients. For team viability, n = 31; for team performance, n = 30.

* $p < .10$

[†] $p < .05$

placed on individual team members. Further, frequent interactions with critical external parties regarding project scope and task requirements should reduce task uncertainty. Team members who face less uncertainty with respect to their task are better able to effectively prioritize work loads, are less likely to expend effort inefficiently, and are therefore less likely to experience overload. Lower task uncertainty is also likely to result in lower general anxiety among team members. Additionally, boundary-spanning activities may help team members form coherent shared mental models of their team's external environment and requirements, which may facilitate greater coordination of team members' actions in interfacing with their environment and so ease experienced role overload. As Choi (2002) noted, there may be synergistic gains between external and internal team activities. Should internal team functioning improve as a result of team boundary spanning, these process benefits will reduce inefficiencies as a team completes its task and will alleviate confusion within the team. Both of these effects should reduce role overload.

The cross-level finding also contributes to broader role theory literature by offering greater theoretical precision for understanding the impact that boundary spanning has on team members. Earlier research identified role stressors associated with boundary spanning (Beehr et al., 1976; Kahn et al., 1964), but our study suggests it is necessary to simultaneously examine the influence of team context on members' potential role overload, highlighting an important contingency for previously well-established findings. As organizations increasingly rely on teams to span functional and organizational boundaries, teams will face even greater external dependencies, and team members, in turn, will be expected to assume additional boundary-spanning responsibilities (Ancona et al., 2002; Choi, 2002). Therefore, it becomes ever more critical to understand how boundary-spanning efforts influence role overload. Our findings further highlight this need, showing that role overload—when viewed in the aggregate—is not only costly to team members, but also detrimental to team viability.

By demonstrating a positive relationship between team-level boundary spanning and team viability (both a direct relationship and an indirect one via reduced role overload), we also provide theoretical insights into an emerging question in research on team boundary spanning regarding the interplay between external and internal team dynamics. Ancona's (1990) study revealed that teams with the highest levels of external boundary spanning were less cohesive than teams with minimal external activity; however, over time, as the teams continued to interact successfully with their environments, internal processes improved. Choi (2002) developed these ideas further, noting that internally directed and externally oriented processes (including boundary spanning) can be both competing and complementary sets of activities. Although tentative, our findings suggest that encouraging as many team members as possible to engage in boundary spanning may enable teams to maximize both their boundary-spanning function and their internal team functioning. This finding contributes to, and perhaps partially contradicts, earlier studies on "gatekeepers" and other liaison roles at R&D facilities (e.g. Tushman, 1977), which indicated that a few individuals generally performed boundary spanning. Interestingly, however, projects with more complexity and uncertainty consistently had more boundary spanning roles than projects with less complexity and uncertainty.

Our second set of core findings provides an initial understanding of the multilevel mechanisms that support engagement in boundary spanning. Thus, we begin to answer calls to directly explore predictors of team boundary spanning (Choi, 2002; Edmondson, 1999). Our findings confirm the influence of role expectations and self-efficacy and are thus particularly relevant for understanding the functionality and effectiveness of externally oriented team member roles. Specifically, we contribute to the teams literature, by showing that roles facilitate critical team processes (Smolek, Hoffman, & Moran, 1999), as well as to role theory, by showing roles to be a mechanism linking members' contributions and team outcomes. This work also explores extent of agreement on boundary spanning's strategic importance as a focal construct of interest, providing evidence that the positive relationship between assuming a boundary-spanning role and engaging in boundary-spanning behavior is enhanced when members agree on the value and relevance of an external focus. Katz and Kahn (1978) asserted that contextual influences such as member agreement affected the role-behavior relationship, yet to our knowledge this is the first study to test and confirm cross-level moderation in support of their theory.

Finally, our findings have implications for the organizational innovation, learning, and knowledge management literatures, which have noted the role of boundary spanning in the sharing, transfer, and development of knowledge (Argote et al., 2003; Hansen, 1999). By identifying and revealing a potential solution to the boundary spanning paradox, our study suggests that future work in these domains should address how organizations can best manage the boundary-spanning activities of their teams that ultimately contribute to organizational outcomes such as knowledge transfer, organizational learning, and innovation. More generally, our model and its related findings may help to provide the foundation for a multilevel theory of boundary spanning linking the individual, team, and organizational levels.

Practical Implications

This study also offers practical implications for organizations that rely on externally dependent work teams and provides insights into how organizations can better design and support these types of teams. First, our findings suggest that, rather than assigning specific team members to boundary-spanning roles, building boundary spanning into the responsibilities of all team members will better enable a team to maximize its boundary spanning function while also mitigating personal costs to team members that, if left unchecked, will negatively impact the team's ability to function productively over time. Further, to mitigate the potential for diffusion of responsibility to occur with respect to boundary spanning, it is important that organizations develop norms of collective accountability that hold everyone, rather than just a single individual, accountable for this important function and explicitly encourage teams to view boundary spanning as a strategic priority. Second, in showing that boundary-spanning self-efficacy predicts team members' engagement in the focal behaviors, our results suggest it is important to take steps to develop team members' efficacy for taking on boundary-spanning activities by providing teams with encouragement and exposure to successful examples and role models (cf. Bandura, 1997). Third, by showing that a team's external orientation relates positively to individual boundary spanning and, if shared, strengthens the relationship between boundary spanning and role behavior, our findings imply that organizations should help teams to understand and learn how to manage their external dependencies. To this end, organizations might actively encourage team training and development activities as well as planning sessions in which team members can openly discuss and create a common, shared understanding of the value and importance of the boundary-spanning function, translate this into externally oriented strategies, and formulate action plans.

Limitations and Directions for Future Research

The current findings must be considered in light of the study's strengths as well as its limitations, which point to important directions for future research. Regarding sample considerations, the consulting teams in this setting were comprised of full-time MBA students, who may differ in motivation, knowledge, and skills from full-time employees. In addition, this was not an ideal setting for studying external leadership, as the external leaders here were faculty advisors who played a minimal leadership role and were substantially less directive and less involved than typical team managers. Further, the primary boundary-spanning concern for the teams in our sample was with external clients, and the teams had full discretion to decide how to best meet their boundary-spanning needs. Therefore, the findings may not generalize to other team settings in which the nature of team boundaries differ and/or where leader directives or formal role assignments restrict member behaviors. However, even with these differences, these MBA teams are authentic examples of externally dependent work teams and provide a realistic picture of professional consulting teams working in organizations.

With respect to measurement and design, hypotheses involving team boundary spanning and team role overload predicting team viability were tested cross-sectionally. However, the potential for common source bias was mitigated, since each of these variables comprised ratings averaged across members and utilized a different referent, and the viability measure reflected summative ratings made over the large majority of each project. Similarly, the testing of Hypothesis 1 was cross-sectional, although again, our use of different referents mitigated common source bias, and strong theory supports a causal relationship between roles and associated role behavior (Katz & Kahn, 1978). Finally, we did not test the stability over time of study variables such as external focus.

Although our results tentatively suggest that high levels of boundary spanning by all team members is ideal, alternative configurations are possible, such as boundary spanning by team members with the highest performance (e.g., measured as a maximum score [Steiner, 1972]). Related alternative configurations might further explain the experience of role overload. For instance, in performing a critical boundary-spanning task that is "disjunctive" (that is, only one member needs to perform it particularly well), other team members also engaging in boundary spanning may feel as if they are performing extra-role activities and therefore may experience more role overload. In contrast, performing a "conjunctive" task might reduce overload through generating an expectation that boundary spanning is the responsibility of all team members rather than an extra-role activity. Another interesting question is how team members might best share boundary-spanning roles. Future work exploring these compositional issues will enrich understanding of the ways in which team members carry out the boundary-spanning function.

Regarding model specification, we suggest that additional antecedents, moderators, and outcomes at higher levels of analysis might be explored. Further, our study did not include internal team process variables, such as coordination and interpersonal dynamics (Marks et al., 2001), which could be antecedents or moderators of the relationship between role overload and team outcomes. Finally, more exploration of antecedents to boundary spanning would also be fruitful. For instance, functional breadth (Bunderson & Sutcliffe, 2002) might predict boundary-spanning behavior, as individuals with multifunctional backgrounds may be more adept at connecting and communicating with a variety of external parties (Choi, 2002). Additionally, other team and contextual factors, such as task complexity, team composition, temporal shifts of external demands and interdependencies (Choi, 2002), and external team leadership style (Ancona, 1990; Druskat & Wheeler, 2003) are likely to have strong implications for member boundary spanning.

Lastly, although our design did not allow us to directly investigate organization-level outcomes, given that teams are an important means of knowledge transfer (Argote et al., 2003), a promising extension of this research would be to investigate how the factors identified in this study at the individual and team levels contribute to boundary spanning capabilities that, at the organizational level, are known to influence internal and external knowledge transfer, which ultimately translates into innovation. Similarly, our findings can be extended to the organizational level in the case of multiteam systems that require high levels of coordination over component teams (Marks et al., 2005) by investigating whether team boundary spanning facilitates interteam coordination in ways that promote the effectiveness of multiteam systems.

Conclusion

This study explored a paradox that emerges when recent research on the team-level benefits of boundary spanning is integrated with earlier work demonstrating the overload experienced by those carrying out boundary spanning. Bridging paradigms and levels, we offer a rare examination of trade-offs between behaviors that benefit a team, but also impose personal costs on its members. Interestingly, our results reveal that team and individual characteristics jointly facilitate engagement in critical boundary-spanning behaviors. Carrying out these efforts is indeed taxing for a team's members, yet high levels of boundary spanning within the team can mitigate these personal costs, thus enhancing the team's ability to function productively over time. We hope our study provides needed conceptual and empirical advancement for the boundary spanning and teams literatures and offers an exemplar for future research designed to identify and explore paradoxes such as these within organizational research.

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